

Open Platform for Network Functions Visualisation (OPNFV)

Open Platform for NFV is a collaborative open source platform started by the Linux Foundation in 2014. It facilitates the deployment and evolution of NFV components across various open source ecosystems. Via system level integration, deployment and testing, OPNFV creates a reference NFV platform to accelerate the transformation of enterprise and service provider networks for creating an open source platform that speeds up the development and deployment of NFV.

The objectives behind the development of OPNFV are to create an integrated and verified open source platform for NFV functionality, provide proactive cooperation of end users to validate OPNFV needs, and contribute to and engage in open source projects.

Features

- Provides automated deployment tools, robust continuous integration and global test infrastructure. It enables developer collaboration and rapid integration across the cloud, SDN and cloud ecosystems.
- Provides a set of scenarios that accelerate time-to-market for the development and deployment of NFV.
- Ensures interoperability in deployment, network integration and VNF applications.
- Works in collaboration with various SDOs like ETSI NFV ISG, IETF, MEF, TM Forum and others.

Official website: <https://www.opnfv.org>

Latest version: Euphrates; 5.1

Project Floodlight

Project Floodlight is an SDN open source project supported by Big Switch Networks and the SDN community. It has been designed for easy setup with minimal dependencies and to be user friendly for developers. It offers a modular system, making it simple to enhance with a feature set. It supports a broad range of hypervisor-based virtual switches like Open vSwitch.

The Floodlight controller has a set of common functionalities to control and enquire an OpenFlow network, while applications on top of it have different features to solve various user needs over the network.

Floodlight controller works with the OpenFlow protocol to orchestrate traffic flows in an SDN environment. OpenFlow is one of the first and most widely used SDN standards; it defines the open communications protocol in an SDN environment that allows the SDN controller (the brains of the network) to speak to the forwarding plane (switches, routers, etc) for making changes to the network.

The SDN controller is responsible for maintaining all the network rules and provides the necessary instructions to the underlying infrastructure on how traffic should

be handled. This enables businesses to better adapt to their changing needs and have better control over their networks.

Features

- Offers a module-loading system that makes it simple to extend and enhance.
- Easy to set up with minimal dependencies.
- Supports a broad range of virtual and physical OpenFlow switches.
- Can handle mixed OpenFlow and non-OpenFlow networks – it can manage multiple islands of OpenFlow hardware switches.
- Designed to be high-performance, it is the core of a commercial product from Big Switch Networks.
- Offers support for OpenStack (link) cloud orchestration platform.

Official website: <http://www.projectfloodlight.org/floodlight/>

Latest version: 1.2

POX

POX is an open source Python-based platform for communicating with SDN switches either using OpenFlow or the OVSDB protocol. It can function as an OpenFlow controller or switch, and can also be used for writing network software. It is a very popular tool for researching SDN and network application programming. The POX controller comes pre-installed with the Mininet virtual machine. Using the POX controller, systems administrators can control OpenFlow devices like the hub, switch, firewall, etc.

Different parameters can be passed to POX according to real or experimental topologies, thus allowing you to run experiments on real hardware, testbeds or in a Mininet emulator.

Features

- Provides a Pythonic OpenFlow interface.
- Has reusable sample components for path selection, topology discovery, etc.
- Fully compliant and efficient to run in any operating system environment, and comes pre-installed with a Mininet simulator.
- Supports a graphical user interface (GUI) and virtual architecture similar to NOX.
- Provides better performance as compared to NOX written in Python.

Official website: <https://github.com/noxrepo/>

Latest version: 0.2.3

Indigo

Indigo is an open source project that supports OpenFlow on a wide range of physical and non-virtual switch platforms, and is supported under the Apache 2.0 licence. It is based on SwitchLight by Big Switch Networks and is composed of the